# Q01: From Next-Token Prediction to Capabilities

What the objective can explain—and what it cannot

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# The question

**Student question**
How does next-token prediction lead to problem solving, coding, and instruction following?

- Start with the training objective, not a metaphor.
- Separate a plausible mechanism from a demonstrated claim.
- End with an evaluation design.

# The objective is narrow

$$\mathcal{L}(\theta) = -\sum_{t} \log p_\theta(x_{t+1} \mid x_{\leq t})$$

- Predict the next token from prior context.
- No explicit "reasoning," "coding," or "instruction following" loss term.
- A Transformer is one architecture for computing this conditional distribution.

# Why prediction rewards structure

1. Text and code contain reusable regularities.
2. Better representations of these regularities reduce prediction loss.
3. Prompts supply examples and constraints at inference time.

Important: usefulness for prediction is not identical to human-like understanding.

# In-context learning is a concrete clue

- A prompt can change behavior without changing weights.
- Few-shot prompting demonstrated substantial task adaptation in large LMs.
- Induction-head evidence links a particular circuit family to one in-context behavior in controlled models.

Brown et al. (2020); Olsson et al. (2022)

# Training data matters

**Pre-training**

- text, code, examples
- predicts continuations
- learns broad statistical structure

**Post-training**

- preference/instruction data
- shifts favored responses
- does not erase the distinction from pre-training

# Do not overclaim

- A persuasive answer may be wrong.
- A correct answer may be memorized, pattern-completed, or generalized.
- "Emergent" can depend on the metric and threshold chosen.

Capability claims need held-out, contamination-aware evaluation.

# Mini experiment

Sequence: `A B A B A ...`

- Unigram predictor: choose the globally frequent token.
- Context-copy predictor: find an earlier matching context and copy its continuation.
- Compare accuracy on repeated-pattern sequences and on a changed rule.

The toy heuristic illustrates a context-dependent algorithm; it is not a model of all Transformer behavior.

# Takeaway

**Best current answer**
Next-token training can reward representations and context-sensitive computations that make structured text, code, and demonstrations predictable. It provides a route to broad behavior, not a proof of robust reasoning or human-like intelligence.

References: Vaswani et al. (2017); Brown et al. (2020); Olsson et al. (2022); Wei et al. (2022); Schaeffer et al. (2023).